



# Liquid Color Sort – Professional Project Plan



Professional project plan for building and launching an original Android “liquid color sort” puzzle game. This plan turns the project documentation into an execution-ready program with scope, schedule, team, budget, risks, and success metrics. It assumes original code and assets — the CodeCanyon item is a market reference only.

## 1. Executive summary

| Field         | Detail                                                                                   |
|---------------|------------------------------------------------------------------------------------------|
| Project name  | Liquid Color Sort (working title)                                                        |
| Product       | Original Android puzzle game (liquid sorting genre)                                      |
| Platform      | Android (native, Kotlin)                                                                 |
| Monetization  | AdMob banner + interstitial + rewarded; optional remove-ads IAP                          |
| Target launch | ~14 weeks from kickoff                                                                   |
| Team size     | 3–5 people (lean)                                                                        |
| Primary goal  | Ship a polished, store-ready game with 100+ tested levels and a scalable level generator |

**One-line objective:** Deliver a crash-free, monetized, Google Play–approved puzzle game with strong retention foundations within one quarter.

## 2. Objectives and success criteria

### Business objectives

- Launch on Google Play within the planned timeline.

- Establish a monetization loop that does not harm early retention.
- Build a reusable codebase and level pipeline for future reskins/sequels.

## Success metrics (KPIs)

| Metric                       | Target at launch | Target at 30 days |
|------------------------------|------------------|-------------------|
| Crash-free sessions          | ≥ 99%            | ≥ 99.5%           |
| D1 retention                 | ≥ 35%            | ≥ 40%             |
| D7 retention                 | ≥ 12%            | ≥ 15%             |
| Avg. session length          | ≥ 4 min          | ≥ 5 min           |
| Levels completed / user (D1) | ≥ 8              | ≥ 15              |
| Ad fill / impression health  | Stable           | Optimized         |

## 3. Scope

### In scope (v1.0)

- Core gameplay (tap-to-pour, validation, win detection)
- Splash, main menu, level select, gameplay, win popup, settings, no-internet popup
- Undo, redo, hint, add tube, restart
- 100+ tested levels + deterministic level generator
- Sound, music, vibration with persistent toggles
- AdMob banner, interstitial, rewarded
- Local progress saving
- Analytics events
- Phone + tablet layouts

### Out of scope (v1.0 — backlog)

- Cloud save / cross-device sync
- Leaderboards, achievements

- Theme/skin shop
- Daily challenges, event levels
- iOS / cross-platform build
- Remove-ads IAP (optional stretch)

## Assumptions

- One Android engineer available full-time.
- Designer available part-time across phases.
- AdMob + Google Play developer accounts are ready.
- All assets are originally created or properly licensed.

## 4. Team and roles (RACI)

| Role               | Responsibilities                                      | R/A |
|--------------------|-------------------------------------------------------|-----|
| Product owner      | Vision, priorities, sign-off, store listing           | A   |
| Android engineer   | Game logic, UI, ads, build, release                   | R   |
| Game/UI designer   | Visual style, icons, tubes, backgrounds, store assets | R   |
| Level designer     | Generator tuning, difficulty curve, level validation  | R   |
| QA tester          | Test plan, device testing, regression                 | R   |
| Audio (contractor) | Music loop + SFX                                      | C   |

*Lean variant:* one engineer + one designer can cover most roles, with QA shared and audio outsourced.

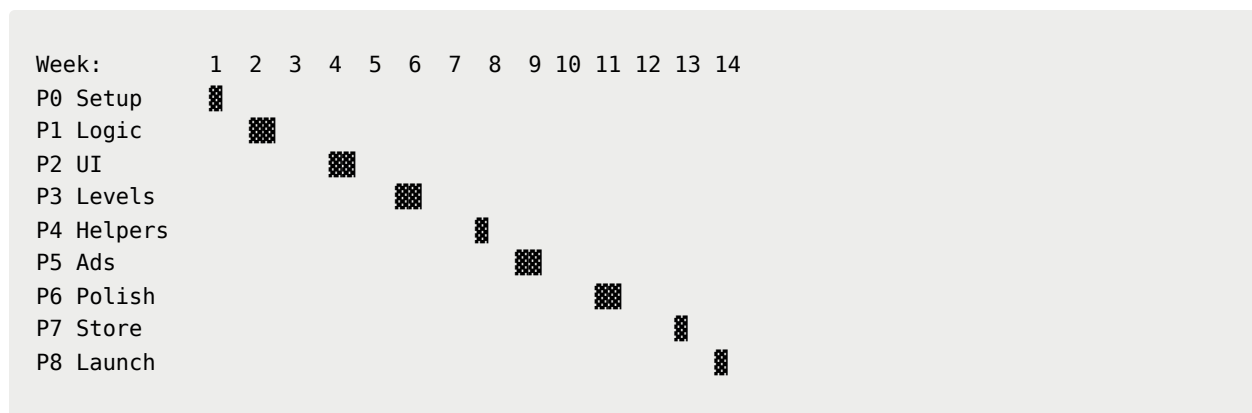
## 5. Timeline and phases

14-week plan organized into 9 milestones. Adjust durations to actual team capacity.

| Phase    | Weeks | Milestone           | Exit criteria                               |
|----------|-------|---------------------|---------------------------------------------|
| P0 Setup | 1     | Project initialized | Repo, package name, CI, navigation skeleton |

| Phase           | Weeks | Milestone         | Exit criteria                                |
|-----------------|-------|-------------------|----------------------------------------------|
| P1 Core logic   | 2–3   | Engine complete   | Pour rules + win detection + unit tests pass |
| P2 Gameplay UI  | 4–5   | Playable build    | Tubes render, tap controls, pour animation   |
| P3 Levels       | 6–7   | 100 levels live   | Generator + level select + progress save     |
| P4 Helpers      | 8     | Assists complete  | Undo/redo/hint/add tube/no-internet          |
| P5 Monetization | 9–10  | Ads integrated    | Banner/interstitial/rewarded with test IDs   |
| P6 Polish       | 11–12 | Feature-complete  | Audio, haptics, animations, tablet layout    |
| P7 Store prep   | 13    | Release candidate | Assets, listing, privacy policy, signed AAB  |
| P8 Launch       | 14    | Live on Play      | Internal test passed, production rollout     |

## Milestone schedule (text Gantt)



## 6. Detailed work breakdown (WBS)

### P0 — Setup (Week 1)

- ☐ Finalize game name + package name
- ☐ Create Android Studio project (Kotlin, Gradle)

- ☐ Set up version control + branching strategy
- ☐ Configure CI build (optional)
- ☐ Build navigation skeleton (splash → menu → game)

## **P1 — Core logic (Weeks 2–3)**

- ☐ Implement Tube, Level, Move, GameState models
- ☐ Move validation ( `canPour` )
- ☐ Pour amount + apply move
- ☐ Win/completed-tube detection
- ☐ Undo/redo stacks
- ☐ Unit tests for all rules

## **P2 — Gameplay UI (Weeks 4–5)**

- ☐ Custom view / canvas tube rendering
- ☐ Tap selection + highlight
- ☐ Pour animation with input lock
- ☐ Move counter + restart
- ☐ Win popup

## **P3 — Levels (Weeks 6–7)**

- ☐ Level data format + loader
- ☐ Deterministic level generator
- ☐ Solvability validation
- ☐ 100+ tested levels
- ☐ Level select screen + locking
- ☐ Progress persistence

## **P4 — Helpers (Week 8)**

- ☐ Hint logic
- ☐ Add tube
- ☐ Undo/redo UI
- ☐ No-internet popup + network util

## **P5 — Monetization (Weeks 9–10)**

- ☐ AdMob SDK integration (test IDs)
- ☐ Banner placement
- ☐ Interstitial frequency capping
- ☐ Rewarded ads for helpers
- ☐ Consent / privacy flow

## **P6 — Polish (Weeks 11–12)**

- ☐ SFX + music + lifecycle handling
- ☐ Vibration/haptics
- ☐ Settings popup persistence
- ☐ Animation polish
- ☐ Tablet/responsive layout

## **P7 — Store prep (Week 13)**

- ☐ App icon, feature graphic, screenshots
- ☐ Store listing copy
- ☐ Privacy policy hosted
- ☐ Production AdMob IDs
- ☐ Signed release AAB
- ☐ Data safety form

## **P8 — Launch (Week 14)**

- ☐ Internal/closed testing
- ☐ Fix blockers
- ☐ Production rollout (staged)
- ☐ Post-launch monitoring

## 7. Deliverables checklist

| Deliverable                  | Phase | Owner          |
|------------------------------|-------|----------------|
| Game design document         | P0    | Product owner  |
| Core engine + tests          | P1    | Engineer       |
| Playable gameplay build      | P2    | Engineer       |
| Level generator + 100 levels | P3    | Level designer |
| Helper features              | P4    | Engineer       |
| Monetization integration     | P5    | Engineer       |
| Final art + audio            | P6    | Designer/Audio |
| Store assets + listing       | P7    | Designer/PO    |
| Signed release build         | P7    | Engineer       |
| Live Play Store app          | P8    | Product owner  |

## 8. Budget estimate (indicative)

Adjust to local rates; ranges are illustrative for a lean indie build.

| Item                          | Type                | Estimate (USD)      |
|-------------------------------|---------------------|---------------------|
| Android development           | Labor / in-house    | Largest cost driver |
| UI/UX + asset design          | Labor / contract    | \$500–\$2,000       |
| Audio (music + SFX)           | Contract / licensed | \$100–\$500         |
| Google Play developer account | One-time            | \$25                |
| AdMob account                 | Free                | \$0                 |
| Tools / plugins / stock       | Misc                | \$0–\$300           |
| QA / test devices             | Optional            | \$0–\$500           |

| Item        | Type          | Estimate (USD) |
|-------------|---------------|----------------|
| Contingency | 10–15% buffer | Variable       |

**Note:** Largest variable is engineering time. If outsourced, scope the engine + UI + monetization as the core paid milestones.

## 9. Risk management

| Risk                        | Likelihood | Impact | Mitigation                                                          |
|-----------------------------|------------|--------|---------------------------------------------------------------------|
| Unsolvable generated levels | Medium     | High   | Generate from solved state; store solution path; solver validation  |
| Ads hurt retention          | Medium     | High   | Delay early ads; cap interstitials; favor rewarded                  |
| Schedule slip               | Medium     | Medium | Weekly checkpoints; protect MVP scope; defer backlog                |
| Color accessibility issues  | Low        | Medium | Distinct palette + optional patterns; device testing                |
| Animation/logic desync      | Medium     | Medium | Single source of truth; lock input; unit tests                      |
| Play Store rejection        | Low        | High   | Original assets; privacy policy; accurate data safety; consent flow |
| Key person dependency       | Medium     | High   | Document code; small modular architecture                           |

## 10. Governance and cadence

- **Weekly checkpoint:** progress vs. milestone, blockers, next-week plan.
- **Phase gate review:** exit criteria sign-off before moving phases.
- **Definition of done:** code reviewed, tested, no known blockers, merged.
- **Change control:** new feature requests go to backlog unless they protect MVP.
- **Launch readiness review:** full release checklist verified before production rollout.

## 11. Post-launch plan (first 30 days)

- Monitor crash-free rate and ANR daily.
- Review retention + drop-off levels via analytics.
- Tune difficulty curve and ad frequency.
- Ship a fast patch for top crashes/bugs.
- Plan v1.1 backlog: more levels, themes, daily challenge, remove-ads IAP.